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# Electromigration in thin aluminum films on titanium nitride

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The aluminum electromigration drift velocity was measured at the temperature range 250–400 °C. A threshold current density was found inversely proportional to the stripe length. An activation energy of 0.55 eV was found for the drift velocity. The occurrence of the threshold is explained by opposing chemical gradients created by the atom pile-up and depletion at the stripe ends. The threshold may explain several observations reported previously. The threshold is increased by decreasing the temperature or by enclosing the aluminum in silicon nitride. Virtually no electromigration is seen for very short aluminum stripes even at current densities above $10^6$ A/cm$^2$.

PACS numbers: 68.60., 66.30.F, 82.45.

## I. INTRODUCTION

Aluminum films are widely used as interconnections on silicon semiconductor devices. A sizable atom flow is induced in such film by passage of high electrical currents. This flow, or electromigration, is parallel to the direction of electron flow. It was found that electromigration can cause device failure due to the opening of the interconnections.[1,2]

The average atom drift velocity $v$ resulting from electrical currents is given by[3]

$$v = J/N = BeZ^*\rho j = (D_0/kT)\, eZ^*\rho j \exp(-\Delta H/kT), \quad (1)$$

where $J$ is the atom flux, $N$ is the density of metal ions, $k$ is Boltzmann's constant, $T$ is the absolute temperature, $eZ^*$ is the "effective change" of the ions, $\rho$ is the metal resistivity, $j$ is the electric current density, $\Delta H$ is the activation energy, $D_0$ is the diffusion constant, and $B$ is the mobility.

It is convenient to rewrite Eq. (1) in the form

$$\frac{vT}{j} = \frac{D_0 eZ^*\rho}{k} \exp\left(-\frac{\Delta H}{kT}\right) \quad (2)$$

which can be used to obtain the activation energy $\Delta H$ and the preexponential factor from the usual Arrhenius plot. Drift velocities were previously obtained by using TEM[4] and SEM[5] techniques. The activation energy, $\Delta H$, was also inferred by resistometric[6] and lifetime[2,7] techniques. Drift velocities are more conveniently and accurately measured by a recently developed technique.[8] This paper describes the results of such measurements on aluminum films deposited on titanium nitride surfaces.

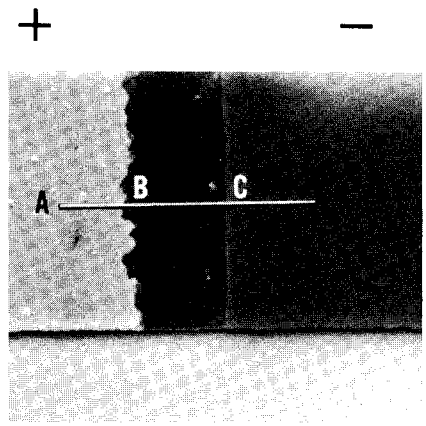
## II. EXPERIMENTAL

$n$-type silicon wafers were oxidized to form a 2000-Å $SiO_2$-type layer. Titanium was reactively sputtered in nitrogen to form a 2600-Å layer of titanium nitride on the oxide layer. The titanium nitride was followed by a 4600-Å aluminum layer, electron beam deposited at a rate of 50 Å/s.

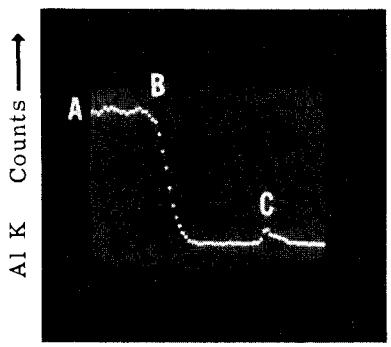
The aluminum and TiN were photolithographically etched to form aluminum stripes 25 $\mu$m wide on TiN conductors. The test structure is similar to that described in Ref. 8.

When electrical currents are passed through the sample, the aluminum stripe carries most of the current since its sheet resistance (~ 0.07 Ω/□) is much lower than that of the TiN (~ 10 Ω/□).

The aluminum atoms move towards the positive terminal, while the aluminum edge facing the negative terminal erodes away. The velocity of this latter edge was assumed to equal the average aluminum drift velocity.[8] The drift of the aluminum edge can be seen in Fig. 1(a). This aluminum film was not annealed prior to powering and shows a relatively clean drifted region.




(a) 10 μm




(b)

FIG. 1. (a) Scanning electron micrograph of the edge of an aluminum stripe after passage of $2.1\times10^6$ A/cm$^2$ for 70 h at 350 °C. The aluminum was annealed 20 h at 350 °C prior to current passage. Aluminum is on the left (A), the drifted region is between (B) and (C), and the TiN is seen to the right of (C). (b) Al$K_\alpha$ x-ray intensity as a function of position along line $\overline{ABC}$. The ordinate is proportional to the intensity.


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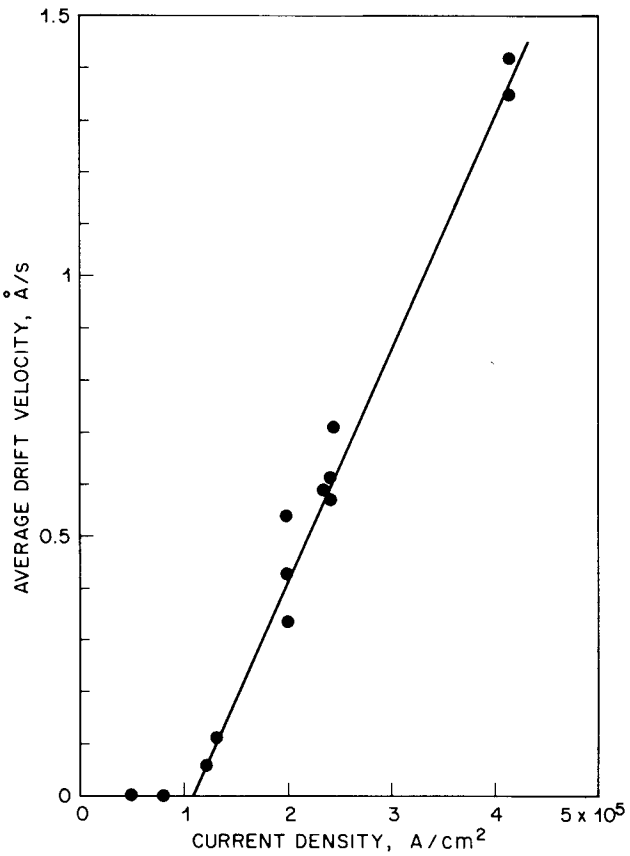



FIG. 2. Average drift velocity of an aluminum stripe (heat treated 500 °C, ½h) as a function of current density. The maximum backflow velocity of the stripe, extrapolated from this graph, is −0.6 Å/s.

Figure 1(b) shows an Al*K* x-ray signal emitted from the sample. There is almost no aluminum remaining in the drifted region.

All experiments were performed in a 85% $N_2$-15% $H_2$ gas mixture. The edge displacement was measured by optical or scanning electron microscope. The samples were preannealed at 500 °C for 30 min resulting in grain growth from about 0.4 to 1—1.5 μm. The drifted region in the 500 °C annealed samples is not as clean as Fig. 1(a) but still permits drift measurements by optical microscopy or by x-ray analysis.

## III. RESULTS

The average drift velocity of the edge of an aluminum stripe 115 μm long was measured as a function of current density at 350 °C (Fig. 2). No drift was detected below a threshold current density of about $1.1\times10^5$ A/cm$^2$; a linear relation between the average drift velocity and current density was found above that threshold.

The slope of the line in Fig. 2 was taken as the appropriate value of $v/j$ in Eq. (2). The reasoning behind this choice will be discussed in Sec. III A. The value of $v/j$ as well as the threshold current density were found to depend on several factors.

### A. Stripe length

The threshold current density was found to be approximately inversely proportional to stripe length. The product of the stripe length $l$ and the threshold current density $j_c$ is approximately constant (for stripe lengths 30 to 150 μm). The threshold-length product for the heat-treated aluminum films, tested at 350 °C, was found to be about $j_c l = 1260$ A/cm.

The effect of stripe length on the average drift velocity can be clearly seen in Fig. 3. Stripes of four different lengths were powered in series, the short stripe did not show any signs of electromigration since it was operating below the threshold, while an increasing amount of drift is seen as the stripe length increases.

The value of the slope, $v/j$, is not a function of length; however, for short stripes (≤30 μm) the velocity was not constant and its deceleration made it impossible to obtain meaningful measurements.

### B. Temperature

Both the threshold-length product and the slope, $v/j$, were found to be a function of temperature. The results of drift velocity measurements at temperatures between 200 and 550 °C are given in Figs. 4 and 5. The drift velocity decreases while the threshold-length product increases with decreasing temperature. The measured activation energy at the low temperature range is about 0.55 eV. If the drift velocity at 200—350 °C is regarded as totally due to grain-boundary electromigration, then the bulk electromigration can be extracted from Fig. 4 by subtracting the extrapolated grain-boundary drift velocity from the total. The bulk Al drift velocity obtained by such subtraction is shown in Fig. 4 and agrees with previous measurements. Previously reported thin-film drift velocity measurements[4,5] are also given in Fig. 4 for comparison.

### C. Thickness

Aluminum films annealed at 500 °C for ½ h were etched to 1000 Å after the annealing. Such films, tested at 350 °C, showed an increase in drift velocity of about 65% over the 4600-Å films. However, the grain structure in the annealed films was such that near the $SiO_2$-

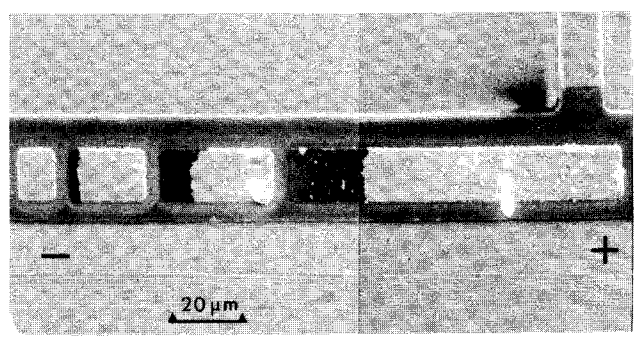



FIG. 3. Drift of four aluminum stripes with varying lengths (heat treated 350 °C, 20 h) after passage of $3.7\times10^5$ A/cm$^2$ for 15 h.

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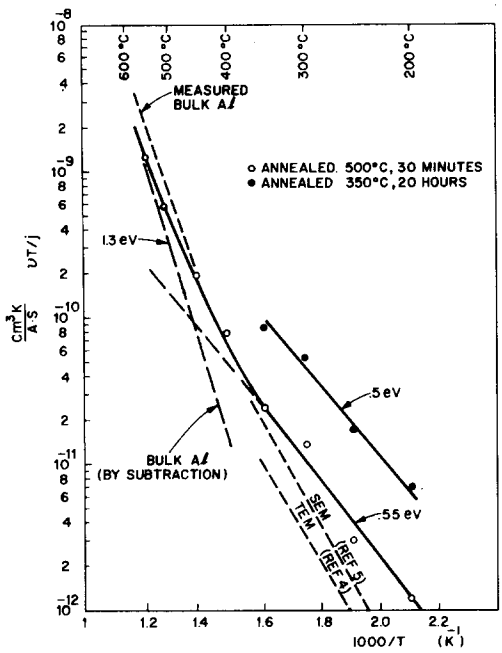



FIG. 4. $vT/j$ vs $1000/T$ for aluminum samples annealed 30 min at 500 °C and 20 h at 350 °C. Samples were 4600 Å thick, 115 μm long, 12 and 25 μm wide.

Al interface there were always more grain boundaries. The observed increase in drift velocity in the thin film may have resulted from the smaller grain size of such films. It is not clear whether surface electromigration is an important diffusion path.

### D. Width

Stripes (annealed $\frac{1}{2}$ h at 500 °C) 115 μm long with widths of 25, 12, 8, and 3 μm were tested at 350 °C. Only the 3-μm-wide stripes showed a lower drift velocity. It was not possible to obtain many measurements with the narrow stripes since they tended to form hillocks and breaks along the entire stripe. It should be noted that this tendency is seen to a lesser degree for nonannealed films. Breaks rarely occur for such films when short stripes are used. When width effects were measured for nonannealed short stripes, the drift velocity deceleration made it impossible to obtain meaningful quantitative values.

### E. Grain size

Films annealed at 350 °C for 20 h having a smaller grain size were tested between 200 and 300 °C. The resulting drift velocities and threshold-length product can be seen in Figs. 4 and 5. Higher drift velocities and lower thresholds were seen for these films. The measured activation energy, 0.5 eV, is somewhat lower than the one for films annealed at 500 °C.

### F. Covering films

The film drift velocity and threshold-length product could also be affected by covering the aluminum with silicon nitride (SiN). It was difficult to observe the location of the stripe edge for annealed films covered with SiN. It was, however, possible to measure the drift velocity for nonannealed thinner films. Figure 6(a) shows such an example, the SiN covering increased the threshold-length product and decreased the drift velocity. However, these effects were seen only when the *entire* aluminum stripe was enclosed within the SiN. A partial covering of the aluminum showed no effect at all on the drift velocity. SiN coverage of wide stripes does not appreciably affect the drift velocity.

## IV. DISCUSSION

The observation that silicon nitride layers did not affect the drift velocity when the stripes were not completely enclosed in them suggested that the retardation of electromigration is related to compressive stress buildup near the anode. This compressive stress effect was already suggested by Ainslie *et al.*[9] as a possible explanation of electromigration retardation in aluminum. The pressure buildup in aluminum was clearly seen in 25-μm-wide stripes where it cracks the overlying SiN. The compressive stress is also capable of forming long aluminum extrusions through holes etched in the cathode region [Fig. 6(b)].

The pressure effects seen for SiN-covered films suggest that similar effects may exist even for uncovered aluminum films possessing only their natural oxide. The electromigration flux transports atoms toward the anode and the compressive stress (as well as the atom concentration) builds up there. This buildup increases the chemical potential of the aluminum to such a level that further transport to the anode is impossible. In other words, the electrical driving force is balanced by a compressive stress gradient which causes an equal but opposing driving force.

This state of events cannot persist for any current density, since there must be a maximum compressive stress that the aluminum film can sustain before yielding. Above this maximum stress, or yield stress, the aluminum deforms plastically which is most commonly manifested in hillock formation and growth. Near the cathode a deficiency in atoms may exist (supersatura-

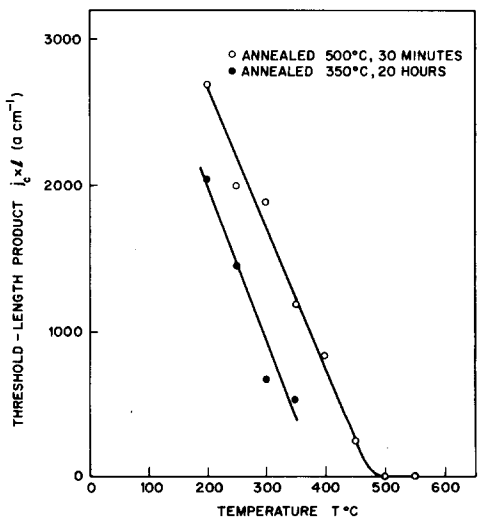
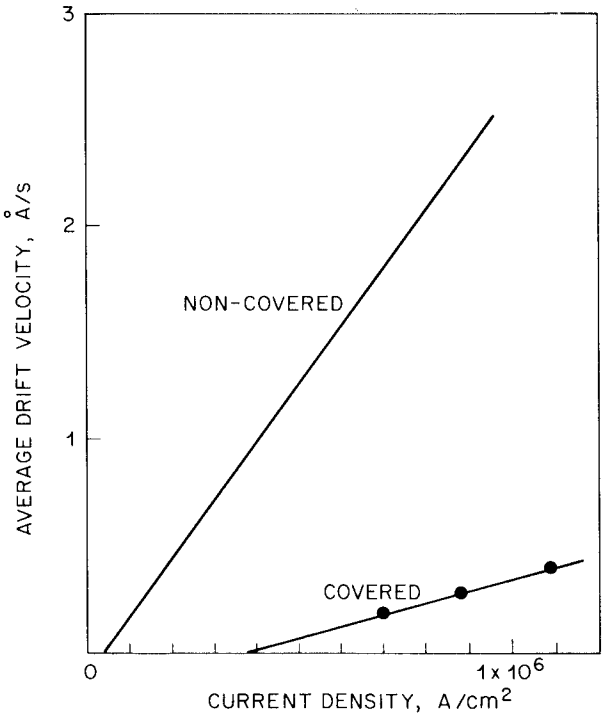
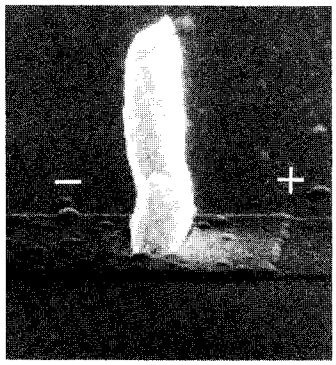



FIG. 5. Threshold-length product for aluminum stripes described in Fig. 4.

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(a)




(b) 10 μm

FIG. 6. (a) Average drift velocity for nonannealed 2000-Å-thick aluminum stripes 115 μm long, 3.7 μm wide at 350 °C covered with SiN and noncovered. The effect of SiN covering should be regarded only qualitatively. Wide variations may occur in the drift velocity. (b) Aluminum extrusion formed through an etched hole in the SiN on the anode end. Current density was $1.5\times10^5$ A/cm$^2$ at 350 °C for 45 h.

tion of vacancies) thus causing a further lowering of the chemical potential. The maximum chemical potential difference, that can build up, divided by the stripe length is then the maximum chemical potential gradient which can oppose the electromigration. Above this maximum, which corresponds to the threshold current density, electromigration will take place transporting atoms along the stripe to form hillocks near the anode. The threshold current density for shorter stripes is higher since larger stress gradients can be created.

The difference in free energy between both ends of the stripe is $\Delta F$. The average gradient in free energy is $\Delta F/l$ (where $l$ is the stripe length). The ensuing back-flow drift velocity is $v_B = B\Delta F/l$. The observed net forward drift velocity, $v_l$, of a stripe is

$$v_l = 0, \quad j \leq j_c \tag{3}$$

$$v_l = v - v_B = BeZ^*\rho j - B\Delta F/l. \tag{4}$$

If the value of $\Delta F/l$ is a constant above the threshold then the value of $v/j$ equals the slope of Eq. (4) since $v/j = BeZ^*\rho j = d\,v_l/dj$. If $\Delta F/l$ does not remain constant then the values of $d\,v_l/dj$ are not equal to $v/j$ but they still remain a good approximation. When $j = j_c$, $v_l = 0$, and by substituting in Eq. (4),

$$j_c l = \Delta F/eZ^*. \tag{5}$$

If it is assumed[9] that $\Delta F \approx \Omega\Delta\sigma_{xx}$ ($\Omega$ is the atomic volume $\Delta\sigma_{xx}$ is the normal stress difference between stripe ends) and any other chemical potential gradients neglected, then Eq. (5) becomes

$$Z^*/\Delta\sigma_{xx} = (j_c l)\,e\rho/\Omega. \tag{6}$$

For heat-treated aluminum films $j_c l = 1260$ A/cm at 350 °C and therefore $Z^*/\Delta\sigma_{xx} \cong 1.2\times10^{-9}$ cm$^2$/dyn. The maximum compressive stress at 350 °C in the aluminum film measured by the bent-beam technique is of the order of $10^9$ dyn/cm$^2$. If such a stress is assumed also in the grain boundaries it follows that $Z^* \approx 1$. This value of $Z^*$ is lower by one or two orders of magnitude from the previously estimated ones.[9,10] The value of $D_0Z^*$ from Fig. 4 is about $10^{-4}$ cm$^2$/s; if $Z^* \approx 1$, then $D_0 \approx 10^{-4}$ cm$^2$/s for the average diffusion preexponential factor and $D \approx 4\times10^{-10}$ cm$^2$/s (for 1-μm grains and boundary width, δ, of 10 Å it follows that the grain boundary-preexponential factor is $\delta D_{ob} \cong 10^{-8}$ cm$^2$/s). The value of the average diffusion coefficient, $D \cong 4\times10^{-10}$ cm$^2$/s, is quite reasonable.[11] Conversely, if it is assumed[10] that $Z^* = 10-100$, then it follows that $D_0 \cong 10^{-5}-10^{-6}$ cm$^2$/s but $\Delta\sigma_{xx} = 10^{10}-10^{11}$ dyn/cm$^2$, a very high value. Equation (6) is based on the equality of the mobilities between the forward and backstreaming atoms. However, if the surfaces provide an effective backstreaming path while being ineffective for electromigration, then it is possible that $Z^*$ is actually much larger than unity while still not creating excessive stresses by electromigration.

Surface tension provides another force acting on the aluminum. The initial movement of the aluminum can be regarded as the nucleation of a void on the Al—aluminum oxide (or Al-TiN) interface. The increase in free energy due to the creation of new interfaces will have a maximum independent of the stripe length. The threshold in this case should not be dependent on stripe length, contrary to the observations.

Equation (6) does not allow us to distinguish between the relative contribution to $\Delta F$ of the anode and cathode regions. In other words, it is not clear what is the relative importance of the aluminum atoms crowding near the anode and the difficulties (if any) in removing atoms from the cathode exposing these new surfaces.

There is some evidence that there cannot be an excessive supersaturation of vacancies in a polycrystalline aluminum.[12] If there were difficulties in nucleation of voids, then the drift velocity should increase once the

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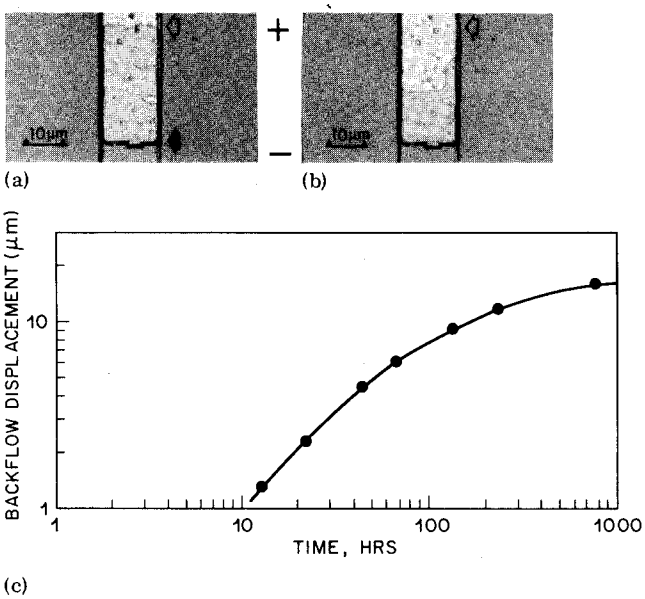



FIG. 7. An aluminum stripe covered with SiN after passage of $4.5\times10^5$ A/cm$^2$ for 19 h at 350 °C. The original edge was at the full black arrow and drifted to the position marked by the open arrow. (b) Backflow of the aluminum after 762 h at 350 °C, no current. (c) Backflow displacement of the edge as a function of time.

voids are formed (if no new aluminum oxide is forming on the newly exposed edge). Actually, in short stripes, a deceleration is frequently seen.

Equation (4) is based on an ideal plastic behavior of the aluminum, i.e., a flow stress independent of strain (and strain rate). The observed deceleration of the drift velocity for very large deformations in short stripes may indeed, in part, reflect an increase in flow stress with large deformation. The effect of strain rate on the flow stress was not studied.

The increase of the threshold-length product with decreasing temperature, seen in Fig. 5, may be related to the increase in the flow stress of the aluminum. At lower temperatures it becomes increasingly difficult to plastically deform the aluminum. Moreover, at low temperatures there could be a tensile stress near the cathode which further increases the stress gradient.

The aluminum backflow can be clearly observed in silicon nitride covered stripes. Figure 7(a) shows the displacement of an edge of a stripe 13 $\mu$m wide after passage of $4.5\times10^5$ A/cm$^2$ for about 19 h at 350 °C. This stripe was connected to a large contact area ($1\times1$ mm$^2$). Heating at 350 °C caused a visible backflow [Fig. 7(b)]. The displacement of the edge as a function of time can be seen in Fig. 7(c). The SiN covering does not affect the nucleation of voids and therefore the decrease in drift velocity due to its presence is most probably related to pressure effects. Short stripes covered with SiN do not show a visible backflow. The excess pressure can be relieved after a very small backflow for the short stripes, whereas stripes connected to big contact areas move a noticeable amount. The latter areas can supply much larger amounts of elastically deformed aluminum, whereas the drifted aluminum in short stripes is all plastically deformed.

## V. CONCLUSIONS

(1) Annealed aluminum films show an electromigration activation energy of ~0.55 eV at the low temperature range.

(2) A threshold current density exists for electromigration. This current is approximately inversely proportional to stripe length. Short stripes can withstand current densities up to $10^6$ A/cm$^2$ without any damage.

(3) The occurrence of a threshold current density may explain several phenomena previously reported in the literature.

(a) It was reported that longer stripes have a shorter time failure for electromigration.[13,14] The lower threshold for longer stripes may be partially responsible for this effect.

(b) It was previously suggested that there might be a threshold in electromigration damage.[15] This threshold must be related to both the threshold in hillock formation and in void nucleation. In fact, the relative contribution of these effects is not known.

In a very long stripe the local damage (hillock and hole formation) is controlled by the average hillock to hole distance. If the current density is below a certain level, the atoms will prefer to continue their movement along the stripe rather than forming a hillock. Holes will not form since they can be readily filled with atoms diffusing from the compressed areas (potential hillock sites). In other words, there is a communication between the potential areas of hole and hill growth. This communication allows a large atom flux in the conductor with a minimum local damage.

(c) The beneficial effect of surface layers[16,17] on electromigration lifetime may be in the forcible increase of this backflow between hill and hole areas. The surface films locally increase the pressure needed to create hillocks thereby eliminating the damage. The compressive stresses needed to suppress local damage are much lower than those calculated by Ainslie[9] since the "effective" strip length is much smaller than its actual length. Long covered stripes can still fail by forming holes near the cathode area. Narrow stripes covered with surface layers are, of course, superior since the surface layer can hold higher compressive stresses in the aluminum without fracturing.

(d) The exponential dependence of the lifetime on current densities, seen in aluminum, may be partially explained by the nonlinear dependence of the drift velocity on current density. Measuring lifetime around the threshold region (the threshold is certainly not as well defined for local damage and may be as high as $10^6$ A/cm$^2$ for closely spaced hillock and hole sites) will give the impression of an exponential dependence on current density.

(4) The lowering of the drift velocity by heat treat-

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ment may be a combination of increased grain size, introduction of impurities into the grain boundaries, and an increase in the yield strength.

(5) Silicon nitride decreases the drift velocity only if it totally encloses narrow aluminum stripes.

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